

Mathematical Foundations for Scientific Thinking

Unit I

Mathematical Language and Logical Foundations

A compact lecture packet with definitions, explanations, worked examples, exercises, and diagnostic questions

Sets · Functions · Logic · Proof · Sequences · Limits

Beryl Meir

Department of Electrical Engineering, Bar-Ilan University

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How to Use This Packet

This unit is meant to be self-contained enough for a technically mature student to read before or after a live lecture. It does not assume that the student has recently studied formal mathematics, but it does assume patience, attention, and willingness to work through definitions carefully.

The order inside each section is deliberate:

1. first the motivation, meaning why the idea exists;
2. then formal definitions and notation;
3. then worked examples with explanations;
4. then check questions and exercises.

The main point of Unit I is not calculation. It is fluency in the language of mathematics. Most advanced technical subjects become easier once the student can read definitions, identify domains, understand quantifiers, and follow proof structure without constant translation.

Learning outcomes

By the end of this unit, the student should be able to:

- read and write basic set notation;
- distinguish a function's domain, codomain, and image;
- negate quantified mathematical statements correctly;
- recognize direct proof, proof by contrapositive, proof by contradiction, and induction;
- work with simple sequences;
- understand the intuitive and formal meaning of a limit.

Sets and Set Operations

Why sets come first

A large part of modern mathematics begins by specifying *what objects are under discussion*. Before one can talk about functions, probability spaces, vector spaces, graphs, manifolds, or solution spaces, one must know what the objects are and when an object belongs to the collection.

A set is the most basic way of doing this. It is a way of drawing a boundary around a collection of objects and saying: these are in, those are out. This sounds simple, but it is already a powerful abstraction. In probability, events are sets of outcomes. In linear algebra, subspaces are sets of vectors. In computer science, databases store sets or collections of records. In analysis, domains are sets on which functions are defined.

Definition: Set

A **set** is a collection of distinct objects, called **elements**. If x is an element of a set A , we write

$$x \in A.$$

If x is not an element of A , we write

$$x \notin A.$$

A set can be written by listing its elements, for example

$$A = \{1, 2, 3, 4\}.$$

It can also be written by describing a property:

$$B = \{x \in \mathbb{N} : x \text{ is even and } x < 10\}.$$

Definition: Common number sets

The standard number systems are:

$$\mathbb{N} = \{1, 2, 3, \dots\} \quad \text{natural numbers,}$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \quad \text{integers,}$$

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\} \quad \text{rational numbers,}$$

$$\mathbb{R} = \text{real numbers,}$$

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}, i^2 = -1\} \quad \text{complex numbers.}$$

Depending on convention, some authors include 0 in $\mathbb{N}$ and some do not. A serious reader always checks the convention being used.

Definition: Subset

A set A is a **subset** of a set B , written

$$A \subseteq B,$$

if every element of A is also an element of B . Formally,

$$A \subseteq B \iff \forall x, x \in A \Rightarrow x \in B.$$

If $A \subseteq B$ and $A \neq B$, then A is a **proper subset** of B , often written $A \subset B$.

Definition: Union, intersection, difference, complement

Let A and B be sets inside some universal set U .

$A \cup B = \{x : x \in A \text{ or } x \in B\},$	union,
$A \cap B = \{x : x \in A \text{ and } x \in B\},$	intersection,
$A \setminus B = \{x : x \in A \text{ and } x \notin B\},$	difference,
$A^c = U \setminus A,$	complement.

The word “or” in mathematics usually means inclusive or: $x \in A$ or $x \in B$ or both.

Worked Example: Basic set operations

Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}.$$

Find $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.

Solution. The union contains everything that appears in at least one set:

$$A \cup B = \{1, 2, 3, 4, 5, 6\}.$$

The intersection contains only the elements appearing in both sets:

$$A \cap B = \{3, 4\}.$$

The difference $A \setminus B$ contains the elements of A that are not in B :

$$A \setminus B = \{1, 2\}.$$

Similarly,

$$B \setminus A = \{5, 6\}.$$

Notice that $A \setminus B$ and $B \setminus A$ are usually different. Difference is not symmetric.

Worked Example: Set-builder notation

Let

$$A = \{x \in \mathbb{N} : x \leq 8\}, \quad B = \{x \in \mathbb{N} : x \text{ is even}\}.$$

Find $A \cap B$.

Solution. The set A is

$$A = \{1, 2, 3, 4, 5, 6, 7, 8\}.$$

The set B contains all positive even integers:

$$B = \{2, 4, 6, 8, 10, 12, \dots\}.$$

The intersection consists of the numbers that are both in A and in B . Therefore

$$A \cap B = \{2, 4, 6, 8\}.$$

The phrase “ $x \in \mathbb{N}$ such that” is not decoration. It tells us the universe in which the property is being evaluated.

Worked Example: Sets in probability

A fair die is rolled once. Let

$$U = \{1, 2, 3, 4, 5, 6\}$$

be the sample space. Define

$$A = \{2, 4, 6\} \quad \text{and} \quad B = \{4, 5, 6\}.$$

Here A is the event “the outcome is even,” and B is the event “the outcome is at least 4.” Then

$$A \cap B = \{4, 6\}$$

is the event “the outcome is even and at least 4.” Also

$$A \cup B = \{2, 4, 5, 6\}$$

is the event “the outcome is even or at least 4.” Since U has six equally likely outcomes,

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}, \quad P(A \cup B) = \frac{4}{6} = \frac{2}{3}.$$

This example shows why set language is not optional in probability. Events are sets.

Check: Sets

1. If $A \subseteq B$, must $A \cap B = A$? Explain.
2. If $A \subseteq B$, must $A \cup B = B$? Explain.
3. Give an example where $A \setminus B \neq B \setminus A$.

Functions and Mappings

Why functions are central

A function is the mathematical language of dependence, transformation, and process. When a scientist writes position as a function of time, voltage as a function of current, energy as a function of state, or output as a function of input, the underlying structure is the same: one object is assigned to another according to a rule.

Equations often get more attention in school, but functions are more fundamental. An equation says that two expressions are equal. A function specifies a relationship from inputs to outputs. Modern mathematics, physics, computation, statistics, and machine learning are filled with functions.

Definition: Function

A **function** from a set X to a set Y is a rule that assigns to every element $x \in X$ exactly one element $y \in Y$. We write

$$f : X \rightarrow Y,$$

and denote the assigned output by

$$y = f(x).$$

The set X is the **domain**. The set Y is the **codomain**. The set of outputs actually reached is the **image** or **range**:

$$\text{Im}(f) = \{f(x) \in Y : x \in X\}.$$

Definition: Injective, surjective, bijective

Let $f : X \rightarrow Y$ be a function.

- f is **injective** or **one-to-one** if different inputs always give different outputs:

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2).$$

Equivalently,

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

- f is **surjective** or **onto** if every element of the codomain is hit by the function:

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y.$$

- f is **bijective** if it is both injective and surjective. A bijection pairs the elements of X and Y perfectly.

Definition: Composition and inverse

If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, the **composition** $g \circ f : X \rightarrow Z$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

If $f : X \rightarrow Y$ is bijective, then it has an **inverse function**

$$f^{-1} : Y \rightarrow X$$

that reverses the action of f :

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(y)) = y.$$

Worked Example: Domain, codomain, and image

Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

The domain is $\mathbb{R}$, because every real number is allowed as an input. The codomain is also $\mathbb{R}$, because the function has been declared as a map into $\mathbb{R}$.

But the image is not all of $\mathbb{R}$. Since $x^2 \geq 0$ for every real number x ,

$$\text{Im}(f) = [0, \infty).$$

This distinction matters. The codomain is part of the function's specification. The image is what the function actually reaches.

Worked Example: Testing injectivity

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = 2x + 1.$$

We test injectivity by assuming two outputs are equal:

$$f(x_1) = f(x_2).$$

Then

$$2x_1 + 1 = 2x_2 + 1.$$

Subtract 1 from both sides:

$$2x_1 = 2x_2.$$

Divide by 2:

$$x_1 = x_2.$$

Therefore f is injective. The logic is: equality of outputs forces equality of inputs, so two different inputs cannot collapse to the same output.

Worked Example: Why x^2 is not bijective on $\mathbb{R}$

Consider again

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

It is not injective because

$$f(2) = 4, \quad f(-2) = 4,$$

but $2 \neq -2$. So two different inputs lead to the same output.
It is also not surjective onto $\mathbb{R}$, because no real number x satisfies

$$x^2 = -1.$$

Thus $-1 \in \mathbb{R}$ is in the codomain but not in the image. Since f is neither injective nor surjective, it is not bijective.

However, if we changed the domain and codomain to

$$f : [0, \infty) \rightarrow [0, \infty), \quad f(x) = x^2,$$

then f would be bijective, with inverse

$$f^{-1}(y) = \sqrt{y}.$$

This shows that the same formula can define different functions depending on the domain and codomain.

Worked Example: Composition

Let

$$f(x) = x^2, \quad g(x) = x + 3.$$

Then

$$(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 3.$$

But

$$(f \circ g)(x) = f(g(x)) = f(x + 3) = (x + 3)^2.$$

In general,

$$g \circ f \neq f \circ g.$$

Composition has order. This is crucial in linear algebra, algorithms, differential equations, and transformations.

Check: Functions

1. Can two different functions have the same formula? Give an example.
2. Is $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3$ injective? Is it surjective?
3. If $f : X \rightarrow Y$ is bijective, what does the inverse function do?

Logic and Quantifiers

Mathematics as disciplined language

A mathematical statement must have a clear truth value. It must be possible, at least in principle, to say whether it is true or false. This is one reason that technical writing is so careful with words like “for all,” “there exists,” “implies,” and “if and only if.” Tiny changes in logical structure can completely change the meaning.

The goal of this section is not symbolic logic for its own sake. The goal is to be able to read theorems accurately. Many students misunderstand advanced mathematics not because the ideas are impossible,

but because they misread the quantifiers.

Definition: Proposition

A **proposition** is a statement that is either true or false.

Examples:

$$2 + 2 = 4$$

is a true proposition, while

$$7 \text{ is even}$$

is a false proposition. By contrast,

$$x + 3 = 5$$

is not a proposition until the value or allowed range of x is specified.

Definition: Implication and equivalence

An **implication**

$$P \Rightarrow Q$$

means: if P is true, then Q is true. It does not necessarily say that P is true.

An **equivalence**

$$P \Longleftrightarrow Q$$

means both directions hold:

$$P \Rightarrow Q \quad \text{and} \quad Q \Rightarrow P.$$

Equivalently, P and Q have the same truth value.

Definition: Universal and existential quantifiers

The universal quantifier

$$\forall$$

means “for all.” The existential quantifier

$$\exists$$

means “there exists.”

For example,

$$\forall x \in \mathbb{R}, \quad x^2 \geq 0$$

means every real number has a nonnegative square.

The statement

$$\exists x \in \mathbb{R} \text{ such that } x^2 = 2$$

means at least one real number squares to 2.

Definition: Negating quantifiers

The negation of a universal statement is existential:

$$\neg(\forall x \in X, P(x)) \Longleftrightarrow \exists x \in X \text{ such that } \neg P(x).$$

The negation of an existential statement is universal:

$$\neg(\exists x \in X \text{ such that } P(x)) \Longleftrightarrow \forall x \in X, \neg P(x).$$

In words: to disprove “all,” it is enough to find one counterexample. To disprove “there exists,” one must show none exist.

Worked Example: Reading a quantified statement

Consider

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } y > x.$$

This says: for every real number x , there is a real number y larger than it. This is true. Given any x , choose

$$y = x + 1.$$

Then $y > x$.

The choice of y may depend on x . That is the meaning of the order $\forall x \exists y$.

Worked Example: Order of quantifiers matters

Compare the two statements:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } y > x$$

and

$$\exists y \in \mathbb{R} \text{ such that } \forall x \in \mathbb{R}, y > x.$$

The first statement is true: after seeing x , we can choose $y = x + 1$.

The second statement is false: it claims there is one real number y that is larger than every real number x . No such real number exists.

This is one of the most important lessons in mathematical reading: changing quantifier order can completely change meaning.

Worked Example: Negating a statement

Negate the statement

$$\forall x \in \mathbb{R}, x^2 \geq 0.$$

Solution. The negation of a universal statement is existential:

$$\exists x \in \mathbb{R} \text{ such that } \neg(x^2 \geq 0).$$

The negation of $x^2 \geq 0$ is $x^2 < 0$. Therefore the full negation is

$$\exists x \in \mathbb{R} \text{ such that } x^2 < 0.$$

This negated statement is false, which agrees with the fact that the original statement is true.

Worked Example: Implication and its converse

Let P be “ n is divisible by 4” and let Q be “ n is even.” Then

$$P \Rightarrow Q$$

is true: every integer divisible by 4 is even.

The converse is

$$Q \Rightarrow P,$$

which says every even integer is divisible by 4. This is false, because 6 is even but not divisible by 4. A common mistake is to prove an implication and then accidentally assume its converse.

Check: Logic

1. Negate: $\forall n \in \mathbb{N}, n + 1 > n$.
2. Negate: $\exists x \in \mathbb{R}$ such that $x^2 + 1 = 0$.
3. Explain the difference between $P \Rightarrow Q$ and $Q \Rightarrow P$.

Proof Techniques

Why proof matters

A proof is not merely a formal ritual. It is a guarantee that a conclusion follows from assumptions. In experimental science, evidence may support a claim. In mathematics, proof establishes that a claim is unavoidable within the given assumptions.

For a technical researcher, proof fluency has another value: it trains the ability to read assumptions. Many failures in modeling and computation arise when someone applies a theorem without checking its hypotheses. Proofs reveal which assumptions are doing the work.

Definition: Proof

A **proof** is a logically valid argument showing that a statement follows from definitions, axioms, and previously established results.

A typical theorem has the form

If hypotheses H hold, then conclusion C follows.

A proof explains why

$$H \Rightarrow C.$$

Definition: Direct proof

A **direct proof** begins with the assumptions and proceeds by valid steps until the desired conclusion is reached. It has the structure

$$P \Rightarrow \dots \Rightarrow Q.$$

Direct proof is usually the first method to try when the definitions naturally point toward the conclusion.

Definition: Contrapositive

The **contrapositive** of

$$P \Rightarrow Q$$

is

$$\neg Q \Rightarrow \neg P.$$

These two statements are logically equivalent. Therefore proving the contrapositive proves the original implication.

Definition: Contradiction

A **proof by contradiction** assumes the opposite of what one wants to prove and then derives an impossibility. The structure is:

Assume $\neg P$. Derive contradiction. Therefore P .

Contradiction is often useful when the desired statement denies existence, such as “there is no rational number whose square is 2.”

Definition: Mathematical induction

Mathematical induction proves a statement $P(n)$ for every $n \in \mathbb{N}$ by showing:

1. **Base case:** $P(1)$ is true.
2. **Inductive step:** for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k + 1)$ is true.

Then $P(n)$ is true for all $n \in \mathbb{N}$.

Induction is the proof method native to recursive and discrete structures.

Worked Example: Direct proof: even squares

Prove: if n is even, then n^2 is even.

Proof. Assume n is even. By definition of evenness, there exists an integer k such that

$$n = 2k.$$

Then

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, n^2 is twice an integer. Therefore n^2 is even.

Worked Example: Contrapositive: odd square implies odd integer

Prove: if n^2 is odd, then n is odd.

Proof. Instead of proving the statement directly, prove its contrapositive:

if n is not odd, then n^2 is not odd.

For integers, “not odd” means even. So it is enough to prove:

if n is even, then n^2 is even.

This was proved in the previous example. Therefore the contrapositive is true, and hence the original statement is true.

Worked Example: Contradiction: irrationality of $\sqrt{2}$

Prove that $\sqrt{2}$ is irrational.

Proof. Assume, for contradiction, that $\sqrt{2}$ is rational. Then there exist integers p, q with $q \neq 0$ such that

$$\sqrt{2} = \frac{p}{q},$$

where the fraction is in lowest terms. Squaring both sides gives

$$2 = \frac{p^2}{q^2},$$

so

$$p^2 = 2q^2.$$

Thus p^2 is even, so p is even. Write $p = 2r$ for some integer r . Then

$$(2r)^2 = 2q^2,$$

so

$$4r^2 = 2q^2,$$

and therefore

$$q^2 = 2r^2.$$

Thus q^2 is even, so q is even.

We have shown that both p and q are even, contradicting the assumption that p/q was in lowest terms. Therefore $\sqrt{2}$ is irrational.

Worked Example: Induction: sum of first n integers

Prove that for every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Base case. For $n = 1$,

$$1 = \frac{1(1+1)}{2} = 1.$$

So $P(1)$ is true.

Inductive step. Assume $P(k)$ is true for some $k \in \mathbb{N}$, meaning

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Then

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This is exactly $P(k+1)$. Therefore, by induction,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Check: Proof techniques

1. Which proof method would you naturally try for “if n^2 is even, then n is even”?
2. In a contradiction proof, what assumption is made at the start?
3. In induction, why is the base case necessary?

Sequences

Sequences as ordered mathematical data

A set has no intrinsic order. The set $\{1, 2, 3\}$ is the same as $\{3, 2, 1\}$. A sequence, by contrast, is ordered. Its first term, second term, and hundredth term matter.

Sequences are the natural language of discrete time, iterative algorithms, approximations, recursive processes, and limiting behavior. Numerical methods generate sequences of approximations. Algorithms often have sequences of states. Infinite series are built from sequences.

Definition: Sequence

A **sequence** is an ordered list of objects indexed by natural numbers:

$$(a_n)_{n=1}^{\infty} = a_1, a_2, a_3, \dots$$

Formally, a sequence in a set X is a function

$$a : \mathbb{N} \rightarrow X.$$

Instead of writing $a(n)$, we usually write a_n .

Definition: Explicit and recursive definitions

A sequence is given **explicitly** if a_n is written directly as a formula in n , for example

$$a_n = \frac{1}{n}.$$

A sequence is given **recursively** if later terms are defined from earlier terms, for example

$$a_1 = 1, \quad a_{n+1} = 2a_n.$$

Recursive definitions are central in algorithms and dynamical systems.

Definition: Arithmetic and geometric sequences

An **arithmetic sequence** has constant difference:

$$a_{n+1} - a_n = d.$$

It has the form

$$a_n = a_1 + (n - 1)d.$$

A **geometric sequence** has constant ratio:

$$\frac{a_{n+1}}{a_n} = r$$

when $a_n \neq 0$. It has the form

$$a_n = a_1 r^{n-1}.$$

Worked Example: Explicit sequence

Let

$$a_n = \frac{2n+1}{n}.$$

Find the first four terms and identify the long-term behavior.

Solution. Substitute $n = 1, 2, 3, 4$:

$$a_1 = \frac{2(1)+1}{1} = 3,$$

$$a_2 = \frac{2(2)+1}{2} = \frac{5}{2},$$

$$a_3 = \frac{7}{3},$$

$$a_4 = \frac{9}{4}.$$

Rewrite the general term:

$$a_n = \frac{2n+1}{n} = 2 + \frac{1}{n}.$$

As n becomes large, $1/n$ becomes small, so the sequence approaches 2.

Worked Example: Recursive sequence

Let

$$a_1 = 3, \quad a_{n+1} = 2a_n - 1.$$

Find a_2, a_3, a_4 .

Solution. Use the recurrence repeatedly:

$$a_2 = 2a_1 - 1 = 2(3) - 1 = 5.$$

Then

$$a_3 = 2a_2 - 1 = 2(5) - 1 = 9.$$

Then

$$a_4 = 2a_3 - 1 = 2(9) - 1 = 17.$$

The key idea is that a recursive definition is not solved all at once. It is unfolded step by step.

Worked Example: Geometric sequence and growth

Let

$$a_n = 5 \cdot 2^{n-1}.$$

Then

$$a_1 = 5, \quad a_2 = 10, \quad a_3 = 20, \quad a_4 = 40.$$

Each term is twice the previous term. This is geometric growth with ratio 2.

Geometric sequences appear in compound interest, population models, radioactive decay, algorithmic divide-and-conquer, and discrete-time systems.

Check: Sequences

1. Is $a_n = 3n + 1$ arithmetic, geometric, or neither?
2. Find the first four terms of $a_1 = 2$, $a_{n+1} = a_n + n$.
3. Explain why a sequence can be viewed as a function.

Limits

Why limits are the gateway to analysis

Calculus depends on the idea of approaching. A derivative asks what happens to an average rate of change as the time interval approaches zero. An integral asks what happens to a sum as the partition becomes infinitely fine. Infinite series ask what happens to partial sums as more and more terms are included.

The limit is the concept that makes all of this precise. Without limits, calculus remains intuitive but not rigorous. With limits, one can say exactly what it means for a quantity to approach a value.

Definition: Convergence of a sequence

A sequence (a_n) **converges** to a number L if its terms become arbitrarily close to L as n becomes large. We write

$$\lim_{n \rightarrow \infty} a_n = L$$

or

$$a_n \rightarrow L.$$

If no such L exists, the sequence **diverges**.

Definition: Formal ε - N definition

A sequence (a_n) converges to L if for every $\varepsilon > 0$, there exists an integer N such that for all $n \geq N$,

$$|a_n - L| < \varepsilon.$$

Symbolically,

$$\lim_{n \rightarrow \infty} a_n = L$$

means

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } n \geq N \Rightarrow |a_n - L| < \varepsilon.$$

The number ε represents the required tolerance. The number N tells us how far into the sequence we must go to stay within that tolerance.

Definition: Limit laws for convergent sequences

If $a_n \rightarrow A$ and $b_n \rightarrow B$, then

$$\begin{aligned} a_n + b_n &\rightarrow A + B, \\ a_n b_n &\rightarrow AB, \\ ca_n &\rightarrow cA, \\ \frac{a_n}{b_n} &\rightarrow \frac{A}{B} \quad \text{if } B \neq 0 \text{ and } b_n \neq 0 \text{ eventually.} \end{aligned}$$

These laws allow many limits to be evaluated without using the formal definition each time.

Worked Example: A basic convergent sequence

Show that

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Explanation. The terms are

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

They become smaller and smaller. To prove convergence formally, take any tolerance $\varepsilon > 0$. We need

$$\left| \frac{1}{n} - 0 \right| < \varepsilon,$$

which is

$$\frac{1}{n} < \varepsilon.$$

This is true whenever

$$n > \frac{1}{\varepsilon}.$$

So choose an integer $N > 1/\varepsilon$. Then for every $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| < \varepsilon.$$

Therefore $1/n \rightarrow 0$.

Worked Example: Rational expression limit

Evaluate

$$\lim_{n \rightarrow \infty} \frac{2n+1}{n+5}.$$

Solution. Divide numerator and denominator by n :

$$\frac{2n+1}{n+5} = \frac{2 + \frac{1}{n}}{1 + \frac{5}{n}}.$$

As $n \rightarrow \infty$,

$$\frac{1}{n} \rightarrow 0, \quad \frac{5}{n} \rightarrow 0.$$

Therefore

$$\lim_{n \rightarrow \infty} \frac{2n+1}{n+5} = \frac{2+0}{1+0} = 2.$$

The principle is that the highest-order term dominates for large n .

Worked Example: Divergent oscillation

Consider

$$a_n = (-1)^n.$$

The sequence is

$$-1, 1, -1, 1, -1, 1, \dots$$

It does not approach a single number. It keeps oscillating between -1 and 1 . Therefore the sequence diverges.

Divergence does not always mean the terms become infinite. It can also mean the terms fail to settle down.

Worked Example: Using the squeeze idea

Suppose

$$-\frac{1}{n} \leq a_n \leq \frac{1}{n}$$

for every $n \in \mathbb{N}$. Since

$$-\frac{1}{n} \rightarrow 0 \quad \text{and} \quad \frac{1}{n} \rightarrow 0,$$

the sequence a_n is trapped between two sequences that both approach 0. Therefore

$$a_n \rightarrow 0.$$

This is the intuitive content of the squeeze theorem: if a quantity is trapped between two quantities approaching the same limit, it must approach that limit too.

Check: Limits

1. Evaluate $\lim_{n \rightarrow \infty} \frac{3n - 7}{2n + 1}$.
2. Does $a_n = \sin(n)$ converge? Give an intuitive answer.
3. In the ε - N definition, what role does N play?

Consolidated Practice Problems

A. Sets

1. Let $A = \{1, 3, 5, 7\}$ and $B = \{3, 4, 5, 6\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.
2. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{2, 4, 6, 8, 10\}$. Find A^c .
3. Prove from the definition that if $A \subseteq B$, then $A \cap B = A$.

B. Functions

4. For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 + 1$, find the image.
5. Determine whether $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x - 4$, is injective and surjective.
6. Give a domain and codomain for which $f(x) = x^2$ becomes bijective.

C. Logic

7. Negate: $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $y > x$.
8. Negate: $\exists n \in \mathbb{N}$ such that $n^2 = 10$.
9. State the converse and contrapositive of: if n is divisible by 6, then n is divisible by 3.

D. Proofs

10. Prove directly: if a and b are even integers, then $a + b$ is even.
11. Prove by contrapositive: if n^2 is even, then n is even.
12. Prove by induction: $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

E. Sequences and limits

13. Find the first five terms of $a_1 = 1$, $a_{n+1} = 3a_n + 1$.
14. Determine whether $a_n = \frac{5n+2}{n-1}$ converges, and find the limit if it does.
15. Give an example of a sequence that is bounded but divergent.

Mini Diagnostic Test

Suggested time: 45 minutes. Use this test to check whether Unit I is solid enough to continue.

1. Define a subset using mathematical notation.
2. Let $A = \{x \in \mathbb{N} : x \leq 6\}$ and $B = \{2, 4, 6, 8\}$. Compute $A \cap B$ and $A \cup B$.
3. Define domain, codomain, and image of a function.
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^2 - 2$. Is f injective? Is it surjective? Explain.
5. Let $f(x) = x + 1$ and $g(x) = x^2$. Compute $(g \circ f)(x)$ and $(f \circ g)(x)$.
6. Negate the statement: $\forall x \in \mathbb{R}, x^2 + 1 > 0$.
7. Negate the statement: $\exists n \in \mathbb{N}$ such that $n < 0$.
8. State the converse of: if n is divisible by 10, then n is even.
9. Prove directly: if n is odd, then n^2 is odd.
10. Prove by contradiction that $\sqrt{3}$ is irrational.
11. Prove by induction that $2 + 4 + 6 + \cdots + 2n = n(n + 1)$.
12. Find the first four terms of $a_n = \frac{n}{n+1}$.
13. Find the first four terms of $a_1 = 2, a_{n+1} = a_n^2$.
14. Evaluate $\lim_{n \rightarrow \infty} \frac{4n + 1}{2n - 3}$.
15. Explain, in words, what the ε - N definition of convergence says.

Brief Answer Key

Checks

Sets. If $A \subseteq B$, then $A \cap B = A$ and $A \cup B = B$. Example of nonsymmetric difference: with $A = \{1, 2\}$ and $B = \{2, 3\}$, $A \setminus B = \{1\}$ but $B \setminus A = \{3\}$.

Functions. Yes, the same formula can define different functions if domain or codomain changes, for example $x^2 : \mathbb{R} \rightarrow \mathbb{R}$ versus $x^2 : [0, \infty) \rightarrow [0, \infty)$. The map $x^3 : \mathbb{R} \rightarrow \mathbb{R}$ is injective and surjective. An inverse reverses the original map.

Logic. The negation of $\forall n \in \mathbb{N}, n + 1 > n$ is $\exists n \in \mathbb{N}$ such that $n + 1 \leq n$. The negation of $\exists x \in \mathbb{R}$ such that $x^2 + 1 = 0$ is $\forall x \in \mathbb{R}, x^2 + 1 \neq 0$. $P \Rightarrow Q$ and $Q \Rightarrow P$ are different implications.

Proofs. For “if n^2 is even, then n is even,” contrapositive is natural. In contradiction, assume the negation of the desired conclusion. The base case anchors induction.

Sequences. $a_n = 3n + 1$ is arithmetic. The recursive sequence $a_1 = 2$, $a_{n+1} = a_n + n$ gives $2, 3, 5, 8, \dots$. A sequence is a function from $\mathbb{N}$ to a target set.

Limits. $\lim(3n - 7)/(2n + 1) = 3/2$. $\sin(n)$ does not converge. N marks how far into the sequence we must go to guarantee the desired tolerance.

Practice Problems: selected answers

1. $A \cup B = \{1, 3, 4, 5, 6, 7\}$, $A \cap B = \{3, 5\}$, $A \setminus B = \{1, 7\}$, $B \setminus A = \{4, 6\}$.
2. $A^c = \{1, 3, 5, 7, 9\}$.
3. If $x \in A \cap B$, then $x \in A$. Conversely, if $x \in A$, then since $A \subseteq B$, $x \in B$, so $x \in A \cap B$. Hence $A \cap B = A$.
4. Image is $[1, \infty)$.
5. $3x - 4$ is injective and surjective as a map $\mathbb{R} \rightarrow \mathbb{R}$.
6. $f : [0, \infty) \rightarrow [0, \infty)$, $f(x) = x^2$.
7. $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, y \leq x$.
8. $\forall n \in \mathbb{N}, n^2 \neq 10$.
9. Converse: if n is divisible by 3, then n is divisible by 6. Contrapositive: if n is not divisible by 3, then n is not divisible by 6.
10. If $a = 2k$ and $b = 2m$, then $a + b = 2(k + m)$.
11. Contrapositive: if n is odd, then n^2 is odd.
12. Induction: base case $1 = 1^2$; add $2(k + 1) - 1$ to k^2 to get $(k + 1)^2$.
13. $1, 4, 13, 40, 121$.
14. Limit is 5.
15. $(-1)^n$ is bounded but divergent.

Diagnostic Test: brief answers

1. $A \subseteq B$ means $\forall x, x \in A \Rightarrow x \in B$.
2. $A \cap B = \{2, 4, 6\}$, $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$.
3. Domain is allowed inputs; codomain is declared target set; image is outputs actually attained.
4. Not injective because $f(1) = f(-1) = -1$. Not surjective onto $\mathbb{R}$ because values below -2 are not hit.
5. $(g \circ f)(x) = (x + 1)^2$ and $(f \circ g)(x) = x^2 + 1$.
6. $\exists x \in \mathbb{R}$ such that $x^2 + 1 \leq 0$.
7. $\forall n \in \mathbb{N}, n \geq 0$.
8. If n is even, then n is divisible by 10.
9. If $n = 2k + 1$, then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, so odd.
10. Same structure as $\sqrt{2}$: assume $\sqrt{3} = p/q$ in lowest terms, get $p^2 = 3q^2$, show $3 \mid p$, then $3 \mid q$, contradiction.
11. Base $n = 1$: $2 = 2$. Inductive step: $k(k + 1) + 2(k + 1) = (k + 1)(k + 2)$.
12. $1/2, 2/3, 3/4, 4/5$.
13. $2, 4, 16, 256$.
14. 2 .
15. For every allowed error tolerance, eventually all terms of the sequence remain within that tolerance of the limit.